

5. Product Note

Additional Client Problem I Chose

Along with answering support questions, I wanted to address the problem of handling actions safely. In a support system, answering a question and performing an action are different things.

For example, if a user asks:

"Please escalate order ORD-1001 to the support team."

I did not want the agent to immediately perform the escalation — an action like this could be triggered accidentally or without the user fully intending to proceed.

To handle this, I implemented a simple confirmation flow. When the user requests an escalation, the system creates a pending action with an action ID and marks it as `PENDING_CONFIRMATION`. The action is not executed at this stage. The user must explicitly confirm it using a command such as:

Confirm ACT-XXXX

Before executing the action, the system checks that the action exists, is still pending, and belongs to the user or account trying to confirm it. Once an action has been executed, it cannot be confirmed again.

I chose this problem because it adds an important safety layer to the agent, and it demonstrates that the agent can handle workflows — not just retrieve information and generate answers.

What I Would Build Next

If I continued developing ParcelPilot, my first priority would be making the action workflow more production-ready:

1. **Persistent actions and audit history** — store actions in a database (action ID, user, account, action type, status, timestamps, confirmation history) so they remain traceable even if the application restarts.
2. **Better document retrieval and citations** — semantic search, metadata filtering, hybrid keyword + vector search, and reranking, plus precise UI citations linking an answer to the exact section of a policy or agreement, so support users can verify why the agent gave a particular answer.
3. **Real authentication and access control** — replace the predefined user contexts with proper authentication and role-based permissions, since customers should only access their own orders, tickets, and actions while internal support users may need broader role-based access.

4. **Human approval for sensitive actions** — extend the pending-action workflow so high-value service credits, refunds, or contract exceptions require manager review and approval before execution, not just user confirmation.
5. **Conversation context** — session memory so a follow-up like *"What workaround should we recommend?"* after *"Investigate ticket TKT-502"* is understood as referring to that ticket.
6. **Monitoring and evaluation** — track tool usage, failed queries, retrieval quality, incorrect answers, blocked unauthorized access attempts, and user feedback to see where the system needs improvement as usage grows.

What I Intentionally Left Out

I focused this submission on building and demonstrating the core requirements rather than a complete production system. I intentionally left out:

- Full production authentication
- Database-backed persistent action storage
- External carrier API integration
- Real-time shipment tracking
- Advanced conversation memory
- Manager approval workflows and UI
- Large-scale document ingestion
- Background processing
- Production monitoring infrastructure

These would be useful in a real product, but I chose to focus my time on making the core agent workflow correct — structured data access, document retrieval, source precedence, account-level security, and safe action confirmation.

Metric I Would Use

The main metric I would use to measure whether ParcelPilot is useful is **Support Resolution Rate** — how many support requests are successfully handled by the agent without requiring unnecessary escalation to a human support representative.

I would also track:

- Accuracy of responses
- Whether the correct sources were used
- Time taken to resolve a request
- Number of unauthorized access attempts correctly blocked

- Successful completion of actions after confirmation
- User feedback

The most important measure, though, is whether ParcelPilot helps users resolve support issues faster while still providing accurate and safe responses.